



Department of Computer Science and Engineering
Final Examination Fall 2024
CSE 321: Operating Systems

Duration: 2 Hours

Total Marks: 45

Answer the following questions.
 Figures in the right margin indicate marks.

1. CO1	Shoumo implemented an application consisting of 38% serial portion and the application is designed in such a way that during the execution it creates multiple threads at a time in such a way where every thread created by the application gets accessed with a designated kernel thread. I. If the number of cores of a system is decreased from 7 to 4 then calculate the change of the performance of the application. II. Identify with proper explanation about the multithreading model which is being used by the application. Mention the drawbacks of the model.	[2+2]
2. CO2	a) In a cloud-based server system, multiple administrators need to access a restricted server for maintenance tasks. To ensure controlled access, each administrator uses a permission code generated by their security app. When an administrator wants to access the server, they must submit their code to the server's authentication system to request access. The server grants access only if the current number of active sessions is below the maximum allowed limit. If the limit is reached, the request will be put on hold until an active session is closed. I. Identify and explain which synchronization method has been used here? II. Logically detect which synchronization problems can be solved by this method?	[2+1]

	b) Find outputs of the given code: <pre> void *func1(int *n); void *func2(int *n); int shared=75; sem_t s1,s2; int main(){ pthread_t t[2]; sem_init(&s1,0,1); sem_init(&s2,0,0); pthread_create(&t[0],NULL,func1,NULL); pthread_create(&t[1],NULL,func2,NULL); for(int i=0;i<2;i++){ pthread_join(t[i],NULL); } sem_destroy(&s1); sem_destroy(&s2); printf("Final Value of Shared: %d\n",shared); return 0; } void *func1(int *n){ sem_wait(&s1); for(int i=0;i<3;i++){ printf("Current Value of Shared: %d\n",shared); shared=shared-10; } sem_post(&s1); sem_post(&s2); } void *func2(int *n){ sem_wait(&s2); for(int i=0;i<2;i++){ printf("Current Value of Shared: %d\n",shared); shared=shared*2; } sem_post(&s2); } </pre>	[3]
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[5]

c) Kabbya, an operating system developer, was experimenting with process synchronization. He came to know of Peterson's solution to the process synchronization problem. In the original solution, there was a variable named turn and an array named flag. He decided to use an array as the turn variable and implement his solution. The code and specifications of his problem is below:

- There are two processes: P0 and P1.
- P0 will start its execution first
- Each assignment statement takes **5 ms** to execute.
- Each boolean logic statement takes **3 ms** to execute.
- Loop statement takes **2 ms** to execute.
- Context Switch will occur after **15 ms**.
- Critical section contains **3** assignments and **2** logical statements.
- Remainder section contains **2** assignments and **3** logical statements.
- For P0: i=0 and j=1.
- For P1: i=1 and j=0.
- turn[0]=FALSE, turn[1]=FALSE.
- flag[0] = FALSE, flag[1] = FALSE.
- If context switch occurs in the middle of executing a statement, the statement is abandoned and restarted again when the process gets the CPU again.

The code used to solve the problem:

```
do{
    flag[i] = true;
    turn[j] = true;
    while(flag[j] == true && turn[j] ==true);
    //critical section
    turn[i] = false;
    flag[i] = false;
    //remainder section
} while(true);
```

Complete the table below for processes P0 and P1 using **Peterson's solution**.

Process 0: i = 0, j = 1	Process 1: i = 1, j = 0
	

3. CO3	a) If we consider Dynamic Partitioning in memory, it will look like the following snapshot at a certain time where gray portions of the memory represent occupied spaces. Illustrate the processes in the memory with the space requirement of P1=50K, P2=150K, P3=100K, P4=73K, P5=75K, P6=30K and P7=25K (in order) applying worst fit and first fit algorithms. Calculate the external fragmentation if there is any. Explain which algorithm makes the most effective use of memory and justify your answer. <table data-bbox="442 521 1059 573"> <tr> <td>80K</td><td>125K</td><td>200K</td><td>25K</td><td>100K</td><td>130K</td></tr> </table>	80K	125K	200K	25K	100K	130K	[3+3 +1]																				
80K	125K	200K	25K	100K	130K																							
	b) Consider a system with a TLB hit ratio of 75%, and a memory access time of 75 ns. If Effective Access Time is 104 ns, then calculate the TLB access time.	[3]																										
4. CO3	a) In a system the size of the main memory is 32 Bytes and size of each page is 4 Bytes. In a certain moment the CPU generates logical addresses 7, 13, 8, 1, 17 and 5 of a process respectively. Logical memory and page table of the process is given below. <table data-bbox="442 976 587 1406"> <tr><th colspan="2">Logical Memory</th></tr> <tr><td>0</td><td>310</td></tr> <tr><td>1</td><td>320</td></tr> <tr><td>2</td><td>330</td></tr> <tr><td>3</td><td>331</td></tr> <tr><td>4</td><td>421</td></tr> </table> <table data-bbox="684 976 829 1406"> <tr><th colspan="2">PMT</th></tr> <tr><td>0</td><td>4</td></tr> <tr><td>1</td><td>8</td></tr> <tr><td>2</td><td>0</td></tr> <tr><td>3</td><td>15</td></tr> <tr><td>4</td><td>6</td></tr> </table> <table data-bbox="925 976 1059 1406"> <tr><th>Main Memory</th></tr> <tr><td> </td></tr> </table> I. Find corresponding physical addresses of generated logical addresses. If the physical address is not valid then write “invalid”. II. Map user’s view of the main memory.	Logical Memory		0	310	1	320	2	330	3	331	4	421	PMT		0	4	1	8	2	0	3	15	4	6	Main Memory		[6+2]
Logical Memory																												
0	310																											
1	320																											
2	330																											
3	331																											
4	421																											
PMT																												
0	4																											
1	8																											
2	0																											
3	15																											
4	6																											
Main Memory																												

	b) Dipu, an operating system developer, is working on memory management. His task is to implement efficient paging on the system he is working on. But recently, due to not increasing his salary, he planned to sabotage the company. He first implemented the OPT algorithm and then attempted to implement a new algorithm which he named MRU algorithm, where the page that was demanded most recently in the past is replaced by the new page. The main memory of his system has 3 frames and the page reference requests are as follows: 7, 4, 6, 7, 2, 0, 5, 1, 4, 7, 1, 2 Apply both the above algorithms and track the page faults. Calculate the page fault ratio. Compare between these two algorithms and mention which algorithm performs better.	[5+5 +2]
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